A QUASI-COHERENT MODULE WITH NO ASSOCIATED POINTS IS ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. This lemma is the vanishing input used by the cards for lemma-restriction-injective-open-contains-ass (L44225) and lemma-ass-zero's siblings; it enters those assemblies as a navigation pointer.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- A module over a Noetherian ring with no associated primes is zero. Algebra, Lemma algebra-lemma-ass-zero, used at proof lines 183-185. Is the algebra statement behind the converse.
- Associated points on an affine scheme are associated primes. Lemma lemma-associated-affine-open, used at proof lines 185-186. Performs the translation.

Source reference. Original lemma 0587: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6770–6770. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02OK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3519–3519. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then*

$$\mathcal{F} = 0 \Leftrightarrow \mathit{Ass}(\mathcal{F}) = \emptyset.$$

Proof. If $\mathcal{F} = 0$, then $\mathrm{Ass}(\mathcal{F}) = \emptyset$ by definition. Conversely, if $\mathrm{Ass}(\mathcal{F}) = \emptyset$, then $\mathcal{F} = 0$ by Algebra, Lemma 0587. To translate from schemes to algebra, restrict to any affine and use Lemma 02OK. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.